FUEL GAUGE READING IS NOT CORRECT [INSTRUMENT CLUSTER]

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Note
- If there is any vehicle malfunction complaint lodged by a customer, perform FOREWORD [INSTRUMENT CLUSTER] malfunction diagnosis according to the troubleshooting procedure. (See FOREWORD [INSTRUMENT CLUSTER].)

Possible cause
- Instrument cluster malfunction
- BCM malfunction
- Fuel gauge sender unit malfunction
- Open or short circuit in wiring harness between fuel gauge sensor unit and BCM

Diagnostic Procedure

Step	Inspection	Action	
1	**PERFORM CAN MALFUNCTION DIAGNOSIS AND DTC INSPECTION** • Perform the CAN malfunction diagnosis according to the recommendation of FOREWORD [INSTRUMENT CLUSTER] malfunction diagnosis, and if there is no CAN malfunction, perform inspection for DTCs. (See DTC INSPECTION [INSTRUMENT CLUSTER].) • Is the DTC displayed?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [INSTRUMENT CLUSTER].)
		No	Go to the next step.
2	**INSPECTION OF INSTRUMENT CLUSTER** • Perform inspection of the fuel gauge/multi-information display referring to the instrument cluster inspection. (See INSTRUMENT CLUSTER INSPECTION.) • Is the display normal?	Yes	Go to the next step.
		No	Replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)
3	**DETERMINE IF MALFUNCTION CAUSE IS FUEL GAUGE SENDER** • Display the following BCM PIDs using the M-MDS. — FUEL_SEN_M • Verify the voltage in the display monitor while shaking the vehicle. • Does the voltage change?	Yes	Go to the next step.
		No	Go to Step 5.
4	**DETERMINE IF MALFUNCTION CAUSE IS BCM** • Display the following instrument cluster PIDs using the M-MDS. — FUEL_SEN_M • Verify the resistance in the display monitor while shaking the vehicle. • Does the resistance change?	Yes	Replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)
		No	Replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/INSTALLATION.)
5	**INSPECT FUEL GAUGE SENDER UNIT** • Switch the ignition off. • Inspect the fuel gauge sender unit. (See FUEL GAUGE SENDER UNIT INSPECTION.) • Is the fuel gauge sender unit normal?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the fuel gauge sender unit and the BCM. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the mid-wiring harness for a short to power supply. • Repair or replace the wiring harness which has a malfunction. **If there is no common connector:** • Repair or replace the wiring harness which has a malfunction.
		No	Replace the fuel gauge sender unit. (See FUEL GAUGE SENDER UNIT REMOVAL/ INSTALLATION.)